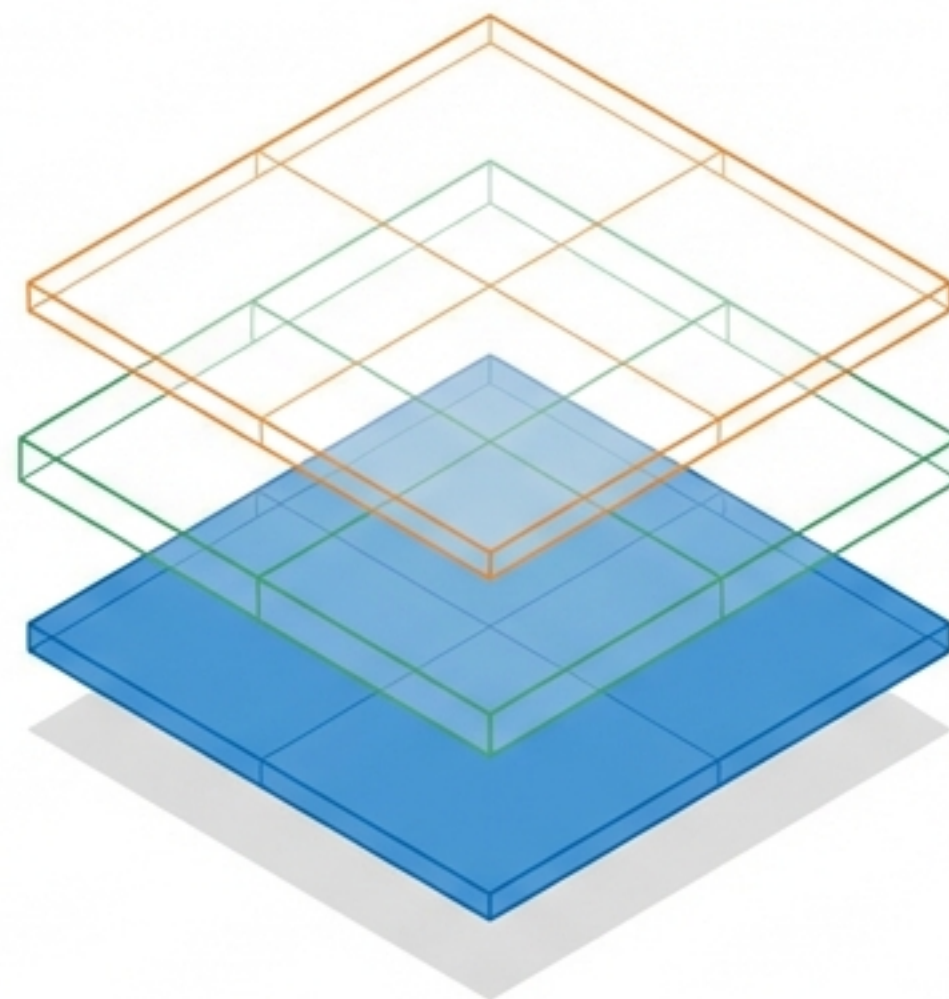




The Three-Layer Architecture

A Separation-of-Concerns Model for the NotebookLM
Infographics & Slides Repository.

DATE:
31 January 2026

AUTHOR:
Dinis Cruz

STATUS:
Approved for implementation

CONTEXT:
Architecture Brief

The 'Flat' Trap: Why the Original Structure Failed



Structural Deficiencies

- **Mixed Concerns:** Raw source files, derived documentation, and publication metadata occupy the same namespace.
- **Navigation Complexity:** Inconsistent depth calculations break breadcrumb logic.
- **Git History Pollution:** Binary assets (images) and text changes are committed together.
- **Rigidity:** The structure cannot accommodate new platforms like Medium or Blogs without breaking links.

Content Has Three Distinct Lifecycles.



1. CREATION (Immutable)

Born on a specific date.
Never changes.
The 'Vault'.

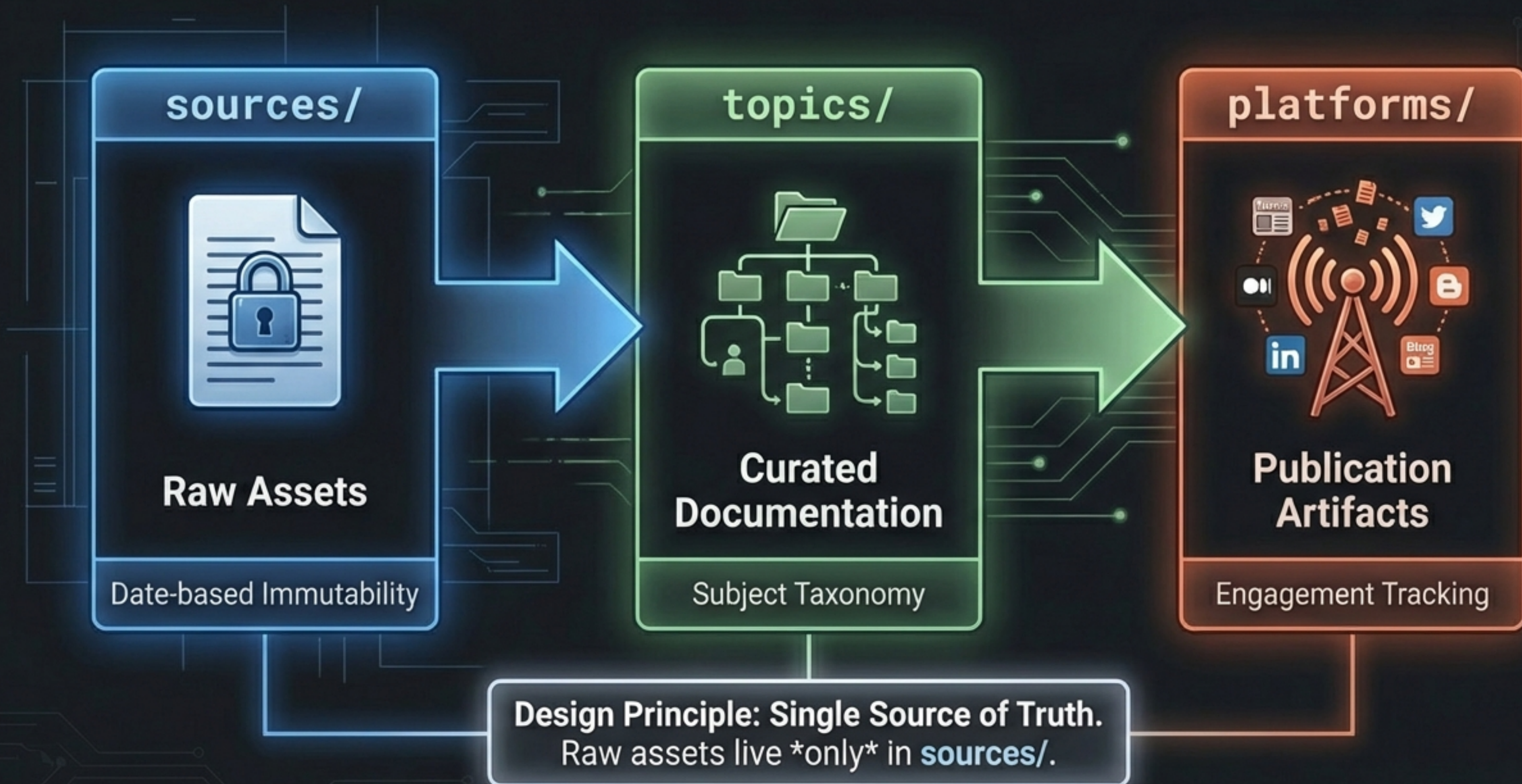
2. CURATION (Evolving)

Organised by topic.
Refined over time.
The 'Library'.

3. PUBLICATION (Ongoing)

Distributed to networks.
Tracks engagement.
The 'Newsstand'.

The Architectural Blueprint.



Layer 1: sources/ – The Immutable Vault.

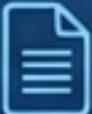
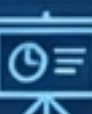
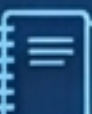
`sources/YYYY/MM/DD/[Topic]/`

Read Only



Read Only

Authorized Artifacts

-  **Source PDF** (Original research/white papers)
-  **Infographic** (NotebookLM generated images)
-  **Slide Deck** (NotebookLM generated PDFs)
-  **Deep Research** (Extended markdown notes)

example file: GDPR implications of deletion.pdf

Naming Conventions for the Source Layer.

File Type	Naming Pattern	Example
Source Document	[Title with spaces].pdf	GDPR implications of deletion.pdf
Infographic	[DD] [Mon] - [Title].jpg	29 Jan - GDPR Compliance.jpg
Slide Deck	[DD] [Mon] - [Title_Underscores].pdf	29 Jan - Gemini_Workspace.pdf

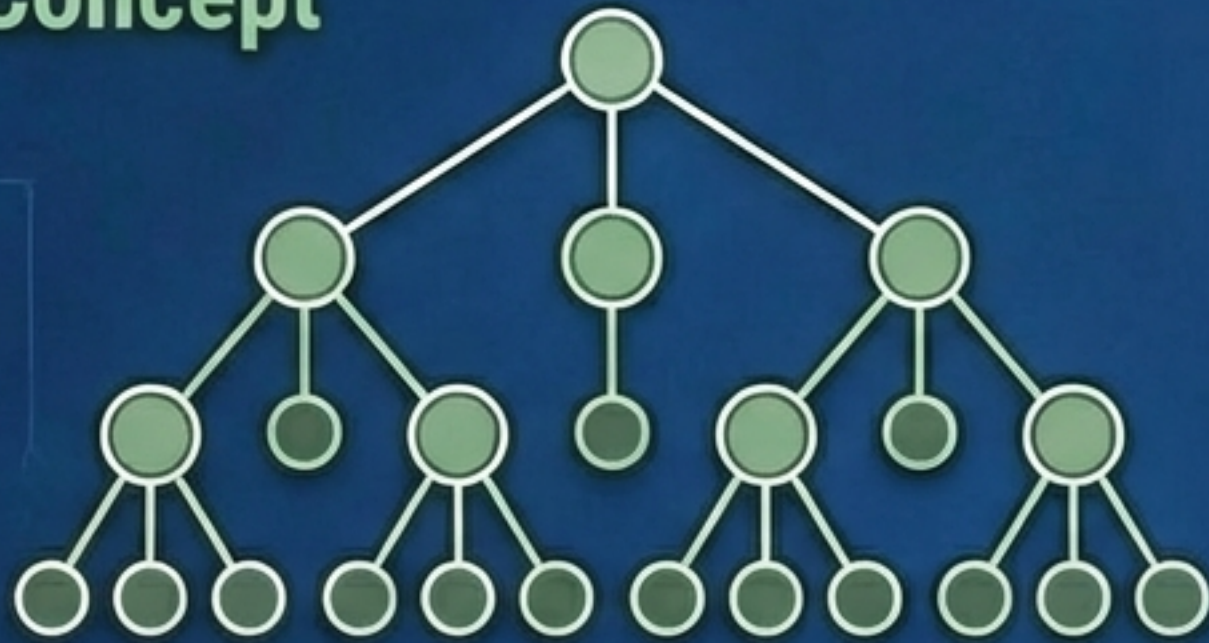
EXCLUDED ITEMS (Do not commit to sources/):

- README.md
- SEMANTIC-GRAPH.md
- slides_mosaic.png
- .webloc files

... topics/Category/Subcategory/Topic/

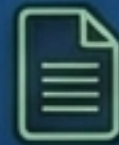
Layer 2: **topics/** – The Curated Library.

Concept

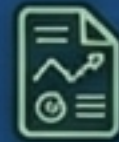


Organised by Domain Concept.

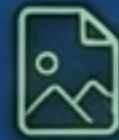
Contents



README.md: The human interface. Contains embedded images and navigation links.



SEMANTIC-GRAPH.md: Machine-readable Knowledge Graph (Mermaid + Cypher).



slides_mosaic.png: A generated 4x4 grid preview of the source slides.

Hierarchy Rule: Group by domain concepts. Subcategorise only when items exceed 20.

Layer 3: **platforms/** — The Publication Engine.

platforms/[platform_name]/Category/Topic/



Why? This layer tracks the 'afterlife' of content once it leaves the repository.

The Mirror Principle: Taxonomy Consistency.

The hierarchy in **platforms/** must strictly mirror **topics/**.

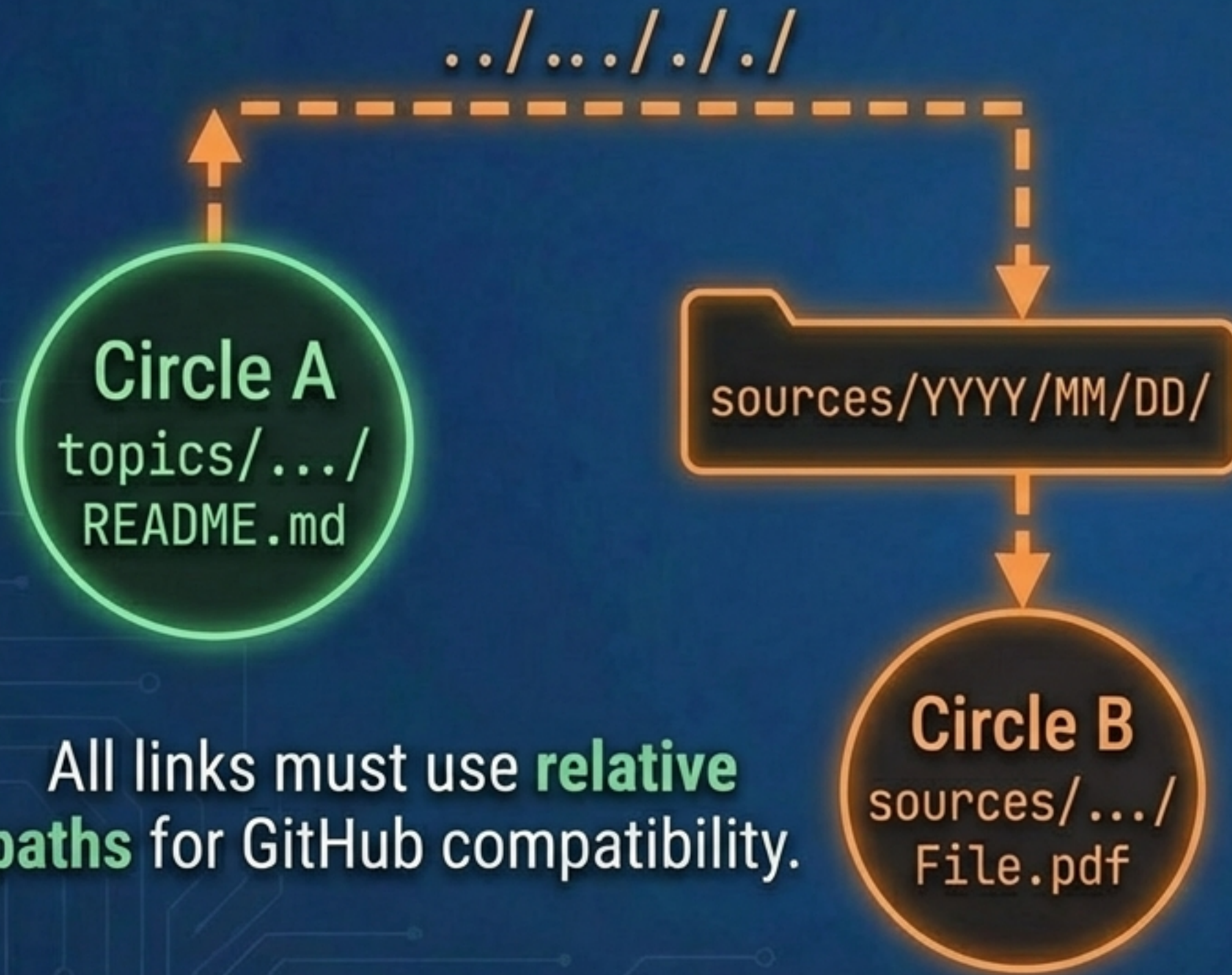


Supported Platforms: LinkedIn (Active), Medium (Future), Blog (Future).

The Separation of Concerns Matrix.

Layer	Lifecycle	Content Function
<code>sources/</code>	Immutable (Creation Date)	The Truth (Raw Assets)
<code>topics/</code>	Evolving (Subject Hierarchy)	The Map (Context & Graphs)
<code>platforms/</code>	Ongoing (Platform + Subject)	The Scoreboard (Links & Metrics)

Connectivity: Relative Linking & URL Encoding.



All links must use **relative paths** for GitHub compatibility.

URL Encoding Standards

Character	Encoding
Space	%20
(	%28
)	%29
En-dash	%E2%80%93

Workflow Part I: Ingest & Curate.

INGEST

Content arrives. Place PDF, Infographic, and Slides in 'sources/YYYY/MM/DD/Topic/'.



CURATE

- Decision: Determine category placement.
- Create folder structure in 'topics/'
 - Generate 'slides_mosaic.png' (pdftoppm)
 - Write 'README.md' and 'SEMANTIC-GRAPH.md'.

Workflow Part II: Publish & Track.

PUBLISH

Action: Post content to LinkedIn.

- Create mirror folder in 'platforms/linkedin/'
- Save '.webloc' file
- Create 'post.md' with copy



UPDATE LOOP

Link & Track.

- Update 'topics/README.md' with link to platform
- Append 'analytics.md' as engagement occurs

Architecture Summary.

Directory	Path Structure	Organization Key	Primary Contents
sources/	sources/YYYY/MM/DD/...	Date Created	Raw PDFs, Decks, Images
topics/	topics/Category/Topic/...	Subject Matter	README, Mosaics, Graphs
platforms/	platforms/Service/Topic/...	Platform Name	Weblocs, Post Text, Analytics

Key Benefits

Clear Separation: Zero ambiguity on where files belong.

Extensibility: Add Blog or Medium later without refactoring history.

Future-Proofing: Analytics and comments live apart from the source of truth.

A Repository That Scales With Knowledge.



“By treating content with the same rigour as code, we create a ‘Code Quality Moat’. This architecture transforms a static archive into a living, navigable engine for insight.”

Architecture approved for immediate migration.